

Netflix Data Analysis Report

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In this project, I analysed movies and TV shows data from the last 50 years using Excel. The process included cleaning the data, creating new columns, and building visualizations to understand trends in ratings, audience engagement, and content patterns.

Dataset Overview

The dataset contains information about movies and TV shows, including details related to release year, ratings, audience engagement, runtime, and age certification. The dataset provides a strong foundation to analyse trends in content production, audience preferences, and performance over time.

Column Feature :

- **Title** - Name of the movie or TV show
- **Type** - Content type (Movie / TV Show)
- **Description** - Short story of content
- **Release Year** - Year of release
- **IMDb Score** - Rating given by users
- **IMDb Votes** - Number of users who voted
- **Runtime** - Duration of the content (in minutes)
- **Age Certification** - Content rating (e.g., PG, R, TV-MA)

Data Preparation

To ensure accurate analysis, the dataset was pre- processed:

- Missing values in **age certification** were handled by assigning “Not Rated”
- Dropped the **imdb_id** column as it not useful in analysis.
- Inconsistent or unclear entries were cleaned
- Data was filtered to focus on the **last 50 years**

Feature Engineering

New columns were created to improve analysis:

- **Last_50_Years** → Identifies records within the last 50 years
- **Age certification clean** → Standardized age categories
- **votes category** → Groups content based on popularity (low / medium / high votes)

Calculated Columns

Last_50_Years - Identifies whether content falls within the last 50 years

Age certification clean - Missing values were replaced with “Not Rated”

votes category - Categorizes content based on number of IMDb votes : Popular and Not Popular

These metrics were created to improve data quality and enhance analysis.

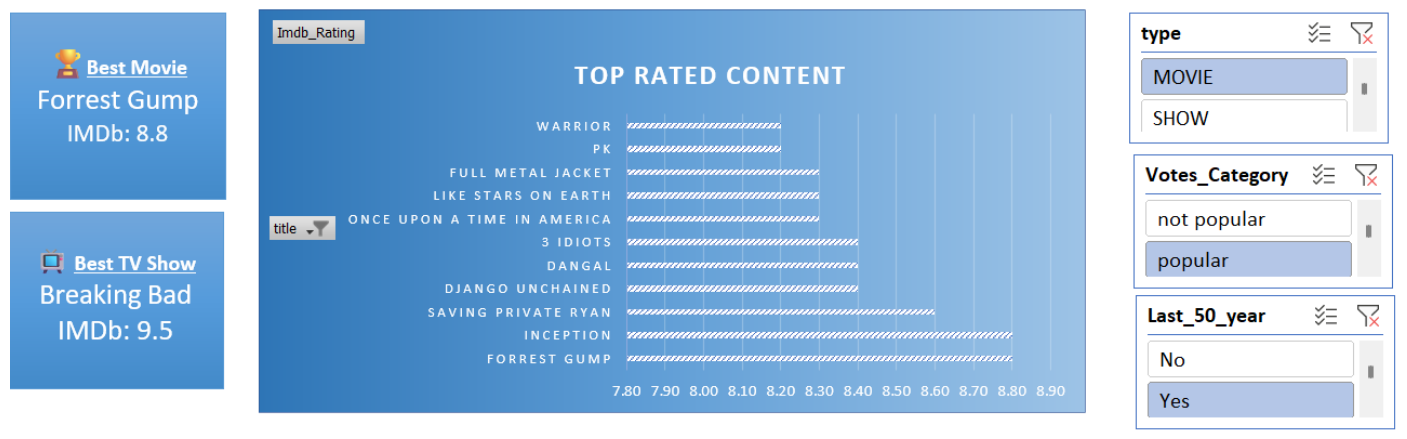
Targeted Questions

1. Which are the highest-rated movies and TV shows over the last 50 years?
2. How many movies are present in the dataset across different years? Is the dataset skewed towards recent years?

- How have IMDb ratings changed over the last 50 years? Have ratings improved, declined, or remained stable?
- Has audience participation (IMDb votes) increased over time?
- How has the average runtime of movies changed over the years?
- How does age certification influence IMDb ratings?

Q1: Best Content Identification

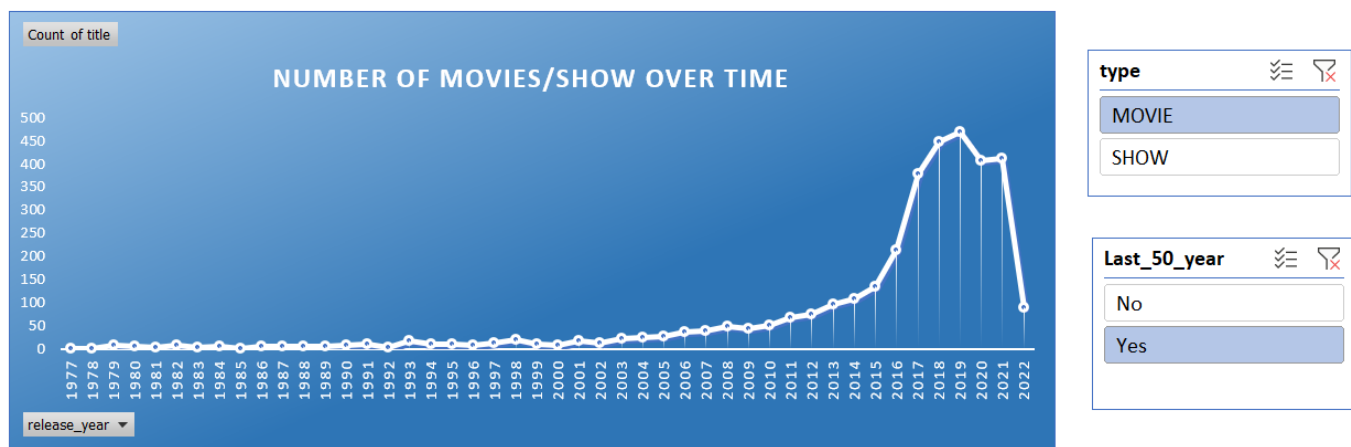
To find the best movie and TV show, I analysed IMDb ratings over the last 50 years. I also considered the number of votes to avoid cases where a title had a high rating but very few viewers. This helped in identifying content that is both highly rated and widely appreciated by audiences.



Q2: Content Distribution Over Time

The analysis shows that the number of movies has increased significantly in recent years, particularly after 2000. Earlier years have relatively fewer entries, indicating that the dataset is not evenly distributed.

This suggests a clear skew towards modern content, meaning the dataset contains more recent movies compared to older ones.



Q3 - IMDb Rating Trend

IMDb ratings were generally higher and more variable in earlier years, especially around the 1980s. Over time, ratings have become more stable but slightly lower on average.

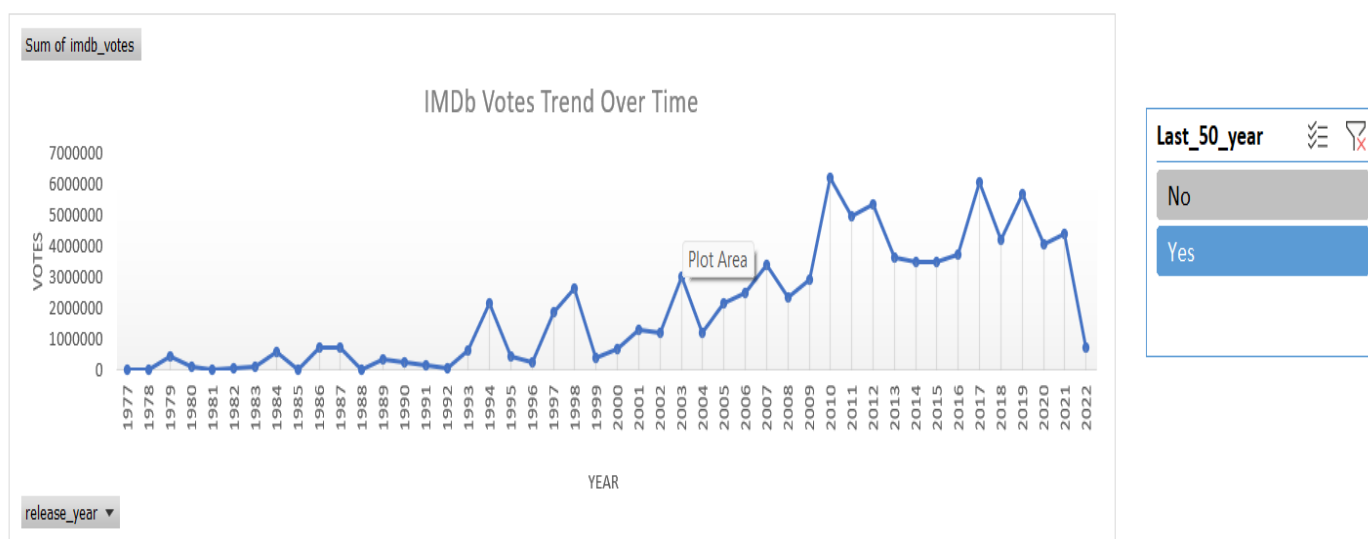
This indicates that while more content is being produced today, average ratings have slightly declined, possibly due to increased competition and content volume.



Q4 - Audience Engagement Trend

The number of IMDb votes has increased significantly over time. Earlier years show very low audience participation, while recent years have seen a sharp rise in voting, reaching into the millions.

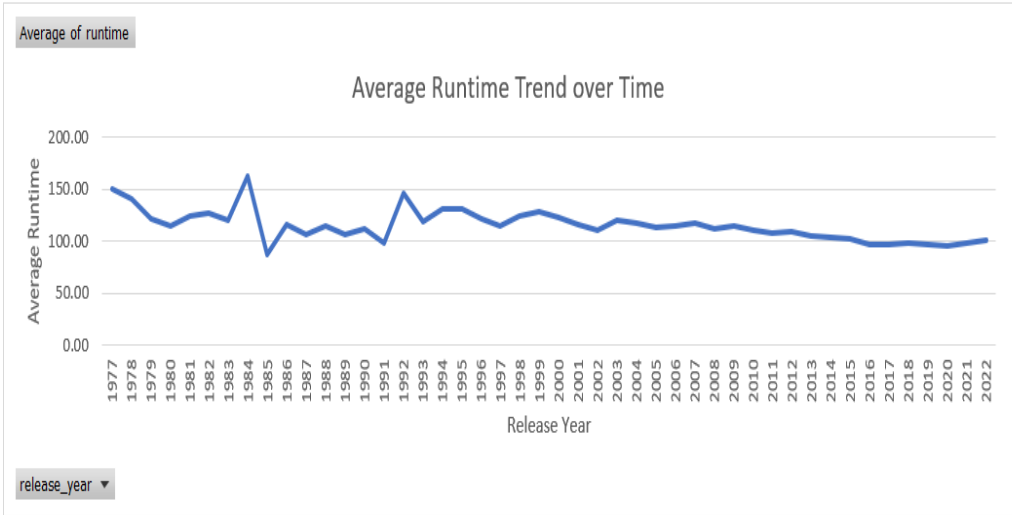
This reflects growing audience engagement, likely driven by increased internet access and the rise of digital and streaming platforms.



Q5 - Runtime Analysis

The average runtime shows a **gradual declining trend over time**. In earlier years (late 1970s and 1980s), movies had longer durations, often exceeding 120–140 minutes. However, over the years, the average runtime has steadily decreased and stabilized around 70–90 minutes in recent years.

This indicates a shift towards **shorter and more concise content**, possibly driven by changing audience preferences and the rise of digital and streaming platforms.



Last_50_year

No

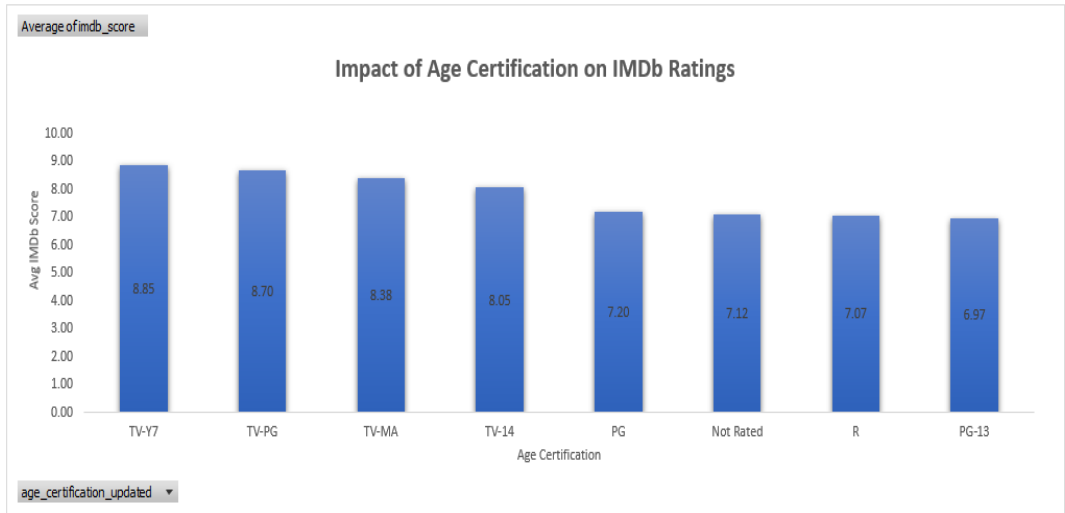
Yes

Q6 - Impact of Age Certification

The analysis shows a clear variation in IMDb ratings across different age certification categories. Content targeted at younger audiences, such as **TV-Y7 (8.85)** and **TV-PG (8.70)**, has the highest average ratings.

As the content becomes more restricted, ratings tend to decline. Categories like **PG (7.20)**, **R (7.07)**, and **PG-13 (6.97)** show comparatively lower ratings.

This suggests that **family-friendly or broadly accessible content tends to receive higher ratings**, while more restricted or niche content may appeal to a narrower audience, resulting in slightly lower average ratings.



type

MOVIE

SHOW

Votes_Category

not popular

popular